

# NIKHIL SUNDA

+91 97999 63897 • nikhilsunda42@gmail.com • linkedin.com/in/nikhil-chdry • github.com/nikhil-chdry

B.Tech (Electrical and Electronics) student at IIIT Bhubaneswar with hands-on experience in full-stack web development, AI/ML engineering, and IoT research. Built production-grade platforms using React, FastAPI, and MongoDB. Published research on smart IoT-based predictive analytics for EV battery systems. Seeking software development, AI/ML, or backend engineering roles.

## Education

### B.Tech in Electrical and Electronics

2027 (Expected)

IIIT Bhubaneswar

CGPA: 7.89 (till 5th Semester)

• **Relevant Coursework:** DBMS, OOPs, Operating Systems, Computer Networks, DSA, NLP

## Skills

**Programming Languages:** C, C++, Python, SQL, JavaScript, Java

**AI / ML:** Machine Learning, Deep Learning, NLP, TensorFlow, PyTorch, Scikit-learn, Pandas, NumPy

**Web Development:** HTML, CSS, React.js, Node.js, Express.js, FastAPI, REST APIs

**Core Concepts:** DSA, OOPs, DBMS, Computer Networks, Operating Systems

**Tools & Engineering:** Git, GitHub, Postman, VS Code, PyCharm, MongoDB, JWT

## Projects

### Algorithmic Smart Shop

GitHub | Live Demo

- Built a full-stack e-commerce platform processing a dataset of 10,000+ products using Python and FastAPI.
- Engineered a **Content-Based Recommendation Engine** using Cosine Similarity to calculate product relationships and display relevant suggestions dynamically.
- Developed a responsive **React** frontend with real-time filtering and optimised **Pandas** pipelines for live data cleaning and validation.
- **Tech Stack:** React, Python, FastAPI, Pandas, Scikit-Learn

### Water AMC Management System (Real-World Business Solution)

GitHub

- Engineered a business management platform for a water treatment company handling RO plants, water coolers, and AMC services.
- Developed scalable backend APIs and responsive frontend interfaces to support daily business operations.
- **Tech Stack:** React.js, JavaScript, Node.js, Express.js, MongoDB, JWT, REST APIs

## Research & Publications

### Smart IoT-Based Predictive Analytics for EV Battery Efficiency and Cooling Systems

Oct 2025

- Published a research paper exploring IoT-enabled predictive analytics techniques for improving electric vehicle battery performance, thermal management, and efficiency.
- Worked on concepts related to EV battery monitoring, cooling optimisation, and intelligent energy management systems.

## Professional Development

### Full-Stack Web Development

- Developing high-performance platforms using **React**, **Node.js**, and **Express** with a focus on **Clean Architecture** and **Glassmorphism UI**.

### AI & Machine Learning

- Engineering **Recommendation Engines** using **FastAPI**, **Scikit-Learn**, and **Cosine Similarity**; exposure to **LLM** workflows and **RAG** concepts.

### Backend & Data Architecture

- Designing **RESTful APIs** and real-time pipelines using **MongoDB**, **Pandas** for data cleaning, and **JWT** for secure authentication.
- Integrated **Leaflet API** for location services and managed **Git/GitHub** workflows for streamlined deployment.

## Extracurricular Activities

---

- Participated in technical and coding events including competitive programming contests.
- Active participant in sports and team-based competitions.
- Engaged in research and innovation activities related to EV technologies and AI applications.